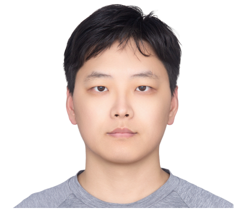
Email: fangyuan.wang@connect.polyu.hk
GitHub: github.com/nicehiro
Homepage: nicehiro.github.io

Fangyuan Wang



I am a third-year Ph.D. candidate in Mechanical Engineering at The Hong Kong Polytechnic University. My research focuses on developing intelligent systems for long-horizon robotic manipulation, with particular expertise in vision-language-action models, subgoal planning, and embodied AI for mobile manipulation tasks.

Research Experience

Human-like Behavior Generation for Humanoid Robots — **Sep 2024 — Present**

Supervised by Prof. David Navarro-Alarcon.
We developed a two-stage cross-embodiment framework for humanoid robots that: (i) generates context-aware, end-effector-constrained human-like poses using advanced perception algorithms, and (ii) retargets motions to diverse humanoid morphologies through innovative shape-adaptive body control mechanisms.

Long-Horizon Manipulation for Robotic Tasks — **Sep 2022 — Present**

Supervised by Prof. David Navarro-Alarcon.

- Designed a two-stage **Vision-Language-Action** pipeline by first predicting proprioceptive subgoal representations, then generating precise joint-space actions for complex manipulation tasks.
- Introduced **Instruction-Augmented Long-Horizon Planning (IALP)**—a closed-loop system that grounds natural-language commands into PDDL via multimodal perception and LLM reasoning.
- Developed **Explicit-Implicit Subgoal Planning (EISP)**—a hybrid framework coupling explicit-implicit subgoal generation to produce feasible, high-value subgoal sequences, enabling robots to solve long-horizon tasks more efficiently.

Byzantine Defense in Decentralized Learning — **Oct 2021 — Sep 2022**

Supervised by Prof. Jian Hou.
We designed a novel credibility-assessment aggregation algorithm that evaluates node reliability through long-term behavioral analysis, achieving 95% accuracy in Byzantine-resilient convergence while maintaining robustness against up to 40% adversarial nodes in fully decentralized learning environments.

Active Flow Control and Optimal Sensor Placement — **Aug 2021 — Sep 2022**

Supervised by Prof. Ning Gui during visiting research at Central South University.
We optimized sensor placement for 2D Kármán vortex street drag reduction using distributed reinforcement learning. We implemented A3C with multi-environment architecture, reducing training time while improving data utilization. We achieved 25% drag reduction by identifying optimal 5-sensor configuration from 151 candidates using attention-based Proximal Policy Optimization.

Multi-Agent Resilient Consensus — **Jul 2020 — Jun 2021**

Supervised by Prof. Jian Hou.
We bridged multi-agent reinforcement learning with resilient consensus by developing adaptive adjacency weight learning via DDPG to automatically identify and reject adversarial neighbors. We also proposed Q-consensus algorithm to eliminate neural network dependency, reducing computational overhead while maintaining convergence guarantees through optimized reward and credibility functions.

Selected Publications

Published Journal Articles

[1] **F. Wang**, et al., "Explicit-Implicit Subgoal Planning for Long-Horizon Tasks with Sparse Rewards," *IEEE Transactions on Automation Science and Engineering (T-ASE)*, vol. 22, pp. 16038-16049, 2025.

[2] J. Wei, **F. Wang**, et al., "Understanding via Exploration: Discovery of Interpretable Features with Deep Reinforcement Learning," *IEEE Transactions on Neural Networks and Learning Systems (T-NNLS)*, vol. 35, no. 2, pp. 1696-1707, 2024.

[3] J. Hou†, **F. Wang**, C. Wei, H. Huang, Y. Hu, and N. Gui, "Credibility-Assessment-Based Byzantine-Resilient Distributed Learning," *IEEE Transactions on Dependable and Secure Computing (T-DSC)*, vol. 20, no. 5, pp. 3734-3746, 2023.

Published Conference Papers

[1] **F. Wang**, et al., "Instruction-Augmented Long-Horizon Planning: Embedding Grounding Mechanisms in Embodied Mobile Manipulation," in *Proc. 39th AAAI Conference on Artificial Intelligence*, Philadelphia, PA, USA, 2025. **(Oral Presentation, 23.4% acceptance rate)**

[2] J. Wei⋆, **F. Wang**⋆, et al., "An Embedded Feature-Selection Framework for Control," in *Proc. 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, Washington, DC, USA, 2022, pp. 4794-4804. **(15.3% acceptance rate)**

[3] J. Hou†, **F. Wang**, L. Wang, and Z. Chen, "Reinforcement Learning Based Multi-Agent Resilient Control: From Deep Neural Networks to an Adaptive Law," in *Proc. 35th AAAI Conference on Artificial Intelligence*, Virtual Event, 2021, pp. 7737-7745. **(21.0% acceptance rate)**

Under Review

[1] S. Lyu⋆, **F. Wang**⋆, et al., "HuBE: Cross-Embodiment Human-Like Behavior Execution for Humanoid Robots," *IEEE Robotics and Automation Letters (RA-L)*, under review.

[2] **F. Wang**, et al., "Think Proprioceptively: Learning Compact Subgoal Representations for Efficient Vision-Language-Action Control," *The Fourteenth International Conference on Learning Representations (ICLR)*, under review.

⋆ Equal contribution   † Supervisor (first student author: **F. Wang**)

Peer Review Experience

- **IEEE Transactions on Neural Networks and Learning Systems (TNNLS)**, 2023–present
- **IEEE Transactions on Robotics (T-RO)**, 2022–present
- **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2025–present
- **IEEE International Conference on Robotics and Automation (ICRA)**, 2022–present
- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)**, 2022–present

Education

Ph.D. in Mechanical Engineering *The Hong Kong Polytechnic University, Hong Kong* Sep 2022 –Present

Dissertation: Long-Horizon Robotic Manipulation with Vision-Language-Action Models

M.Eng. in Software Engineering *Zhejiang Sci-Tech University, China* Sep 2019 –Jun 2022

Thesis: Reinforcement Learning based Multi-Agent Resilient Consensus

B.Eng. in Computer Science & Technology *Zhejiang Sci-Tech University, China* Sep 2014 –Jun 2018